

Gabe Guo

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Email: gabeguo@stanford.edu

Web: gabeguo.github.io

GitHub: github.com/gabeguo

Google Scholar:

<https://scholar.google.com/citations?user=3yn4k9IAAAAJ&hl=en&oi=ao>

Research Interests

Efficient generative modeling

Education

Stanford University

Palo Alto, CA, USA

Ph.D. in Computer Science

09/2024 –

DOE CSGF Fellow

Advisor: Stefano Ermon

Columbia University

New York, NY, USA

B.S. in Computer Science (Intelligent Systems)

09/2020 – 05/2024

Advisors: Hod Lipson, Tony Dear, Richard Zemel

GPA: 4.06/4.33

Selected Publications

Self-Speculative Decoding Accelerates Lossless Inference in Any-Order and Any-Subset Autoregressive Models.

Gabe Guo, Stefano Ermon.

<https://openreview.net/pdf?id=hZnibTOke7>

ICLR 2026.

You Need Better Attention Priors.

Elon Litman, Gabe Guo

<https://arxiv.org/abs/2601.15380>

ICML 2026.

Ab Initio Structure Solutions from Nanocrystalline Powder Diffraction Data

Gabe Guo, Tristan Saidi, Maxwell W Terban, Michele Valsecchi, Simon JL Billinge, Hod Lipson.

<https://www.nature.com/articles/s41563-025-02220-y>

Nature Materials, 2025.

Unveiling Intra-Person Fingerprint Similarity via Deep Contrastive Learning

Gabe Guo, Aniv Ray, Miles Izydorczak, Judah Goldfeder, Hod Lipson, Wen Yao Xu.

<https://www.science.org/doi/10.1126/sciadv.adi0329>

Science Advances 10.2 (2024): eadi0329.

In Progress

ABC: Any-Subset Autoregression via Non-Markovian Diffusion Bridges in Continuous Time and Space.

Gabe Guo, Thanawat Sornwanee, Lutong Hao, Elon Litman, Stefano Ermon, Jose Blanchet.

<https://arxiv.org/abs/2604.27443>

Under review, 2026.

Research Experience

Ermon Group (Stanford University)

Mentor: Professor Stefano Ermon 01-2025 – Present
Researching the intersection of stochastic processes and deep generative models.

Quantum Simulations Group (LLNL)

Mentor: Dr. Keith Ray, Dr. Shivani Srivastava, Dr. Vincenzo Lordi 06-2025 – 09/2025

Led project in using machine learning to analyze two-level systems in amorphous silicon, with application to screening quantum decoherence.

Guibas Group (Stanford University)

Mentor: Professor Leonidas Guibas, Dr. Congyue Deng 03-2025 – 06/2025
Led research project in efficient activation functions for equivariant networks.

Z Group (Columbia University)

Mentor: Professor Richard Zemel 03/2024 – 09/2024
Led research project in probabilistic self-supervised learning with masked transformer autoencoders.

Billinge Group (Columbia University)

Mentor: Professor Simon Billinge 12/2023 – 09/2024
Led research projects in: score-based generative models for nanostructure solutions from x-ray diffraction patterns, generative modeling of electron density fields.

Creative Machines Lab (Columbia University)

Mentor: Professor Hod Lipson 06/2022 – 08/2024
Led research projects in: contrastive learning for cross-finger biometric recognition (patent pending), learning-based configuration space approximation, and deep 3D generative modeling for crystallography.

NSF REU in Biometrics

Mentor: Professor Wenyao Xu 05/2021 – 08/2021
Led deep learning research project in which we were the first in the world to discover the intra-person fingerprint correlation.

Embedded Sensors and Computing Lab (SUNY Buffalo)

Mentor: Professor Wenyao Xu

05/2018 – 04/2021

Various research projects in smart health, ubiquitous computing, and HCI.
Started as a high school student.

Teaching Experience

Head Teaching Assistant, Computer Science (Columbia University)

Mentor: Professor Tony Dear

09/2023 – 05/2024

Created homeworks and exams, coordinated TA team, and organized weekly recitations for COMS 3203 (Discrete Mathematics) and COMS 4733 (Computational Robotics), in addition to typical TA responsibilities.

Teaching Assistant, Computer Science (Columbia University)

Mentor: Professor Tony Dear

09/2021 – 05/2023

Hosted office hours, graded homeworks, and taught recitations. Classes include: COMS 3251 (Computational Linear Algebra), COMS 3203 (Discrete Mathematics), COMS 4701 (Artificial Intelligence).

Community Service

Columbia Engineering Student Council

New York, NY, USA

Vice President

05/2022 – 05/2024

Representative

09/2021 – 05/2022

Organized and led various school spirit events attended by thousands of students. Spearheaded communications to student body. Collaborated with admins to improve schoolwide policies, including expanded fitness center hours, 24-hour dining hall service, and pass-fail grading.

Invited Talks

The New Era of Fingerprint Biometrics: Revealing Cross-Digit Similarity via Deep Contrastive Learning, CSE 709 (Graduate Seminar on IoT and Biometrics), University at Buffalo, Buffalo, NY, USA, November 3, 2023

The New Era of Fingerprint Biometrics: Revealing Cross-Digit Similarity via Deep Contrastive Learning, Seminar on Forensic Science in the Age of AI, New Jersey Association of Forensic Scientists, Newark, NJ, USA, May 10, 2024

Selected Press

CNN

[“Are fingerprints unique? Not really, AI-based study finds”](#)

BBC

[“Our fingerprints may not be unique, claims AI”](#)

BBC Science Focus

[“Your fingerprints aren’t unique after all, discovers AI”](#)

Business Insider

[“A college senior used AI ...”](#)

NPR Science Friday

[“Is Each Fingerprint On Your Hand Unique?”](#)

The Times

[“AI can now link two separate fingerprints from same person”](#)

The Sun

[“AI YOU KIDDING?”](#)

The Register

[“AI flips the script on fingerprint lore”](#)

Metro

[“AI smashes the idea that all fingerprints are unique”](#)

Daily Mail

[“Our fingerprints may NOT be unique, study finds”](#)

<i>EuroNews</i>	“Could AI improve forensics?”
<i>New Scientist</i>	“AI can tell if prints from two different fingers”
<i>Live Science</i>	“Forensic scientists have a new fingerprint-matching tool”
<i>The National Desk</i>	“AI finds new way to pair unmatchable fingerprints”
<i>The Naked Scientists</i>	“AI fingerprint breakthrough could aid future forensics”
<i>Courthouse News</i>	“AI disproves differences between same-hand fingerprints”
<i>The Messenger</i>	“Shocking New Fingerprints Study Could Solve Cold Cases”
<i>Science Alert</i>	“Groundbreaking Study Reveals Your Fingerprints ...”
<i>Futurism</i>	“AI study claims human fingerprints ...”
<i>IFLScience</i>	“AI Overturns Claim That Every Human Fingerprint Is Unique”
<i>Biometric Update</i>	“AI innovates in forensic biometrics”
<i>The Financial Express</i>	“Our fingerprints may not be unique, claims AI”
<i>Business Day</i>	“AI discovers that not every fingerprint is unique”
<i>WION</i>	“US university uses AI to prove our fingerprints are not unique”
<i>WBAY Action 2</i>	“3 BRILLIANT MINUTES: A.I. for doctors and detectives”
<i>iHeartRadio Krime Podcast</i>	“AI Discovers Fingerprints are Not Unique”

Awards

<i>Department of Energy: Computational Science Graduate Fellowship</i>	April 2024
<i>NSF GRFP (Declined)</i>	April 2024
<i>Kosoresow Award for Excellence in Teaching (Columbia CS)</i>	May 2024